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RESEARCH ARTICLE

Adapting Large Language Models for Arabic: Comparative Evaluation, Fine-Tuning, and Ethical Deployment

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ABSTRACT Large language models (LLMs) are reshaping information access, yet adapting them to morphologically rich and resource-scarce languages such as Arabic remains challenging. This paper presents a practical framework for Arabic-focused LLM development and deployment that integrates parameter-efficient fine-tuning (LoRA), retrieval-augmented generation (RAG), and cloud/on-premises strategies. It also introduces an ethics and observability checklist tailored to Arabic contexts. We empirically evaluate three models—TinyLlama-1.1B, Qwen2.5-7B, and DeepSeek-Qwen-14B—on Arabic medical QA and generation tasks using standard text-quality metrics (BLEU-4, ROUGE-L, and BERTScore), factuality and fluency scores, runtime measures (latency, tokens/s), and resource usage. DeepSeek-Qwen-14B achieves the highest accuracy and consistency across specialties. TinyLlama-1.1B delivers the best throughput and smallest memory footprint for edge or latency-sensitive settings, while Qwen2.5-7B provides a balanced trade-off for mid-scale deployments. RAG reduces hallucinations by approximately 12% on knowledge-intensive queries, whereas LoRA fine-tuning improves task accuracy by about 6–9% relative to RAG alone, albeit with additional training cost. We observe slight quality degradation on very long generations—especially for compact models—and variability across medical subdomains, underscoring the importance of domain coverage and dialectal diversity. Overall, the paper contributes a reproducible evaluation pipeline, model-selection guidance, and an ethical deployment checklist, highlighting hybrid RAG + fine-tuning stacks as a practical path to reliable and efficient Arabic LLMs.

INDEX TERMS Generative AI, large language models, Arabic NLP, model deployment, ethical AI, multilingual models, LLaMA, GPT, DeepSeek, Qwen.

I. INTRODUCTION

Generative AI enables machines to produce human-like text, images, code, and other creative content. Since the release of ChatGPT-3 in November 2022, LLMs have rapidly reshaped the global technology landscape, sparking widespread adoption and intense competition among leading companies. Microsoft integrated GPT, DALL·E, and Codex under the name Copilot [1], while Apple introduced GPT-powered functionality as Apple Intelligence [2].

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In 2025, DeepSeek launched DeepSeek-R1, delivering strong performance at a fraction of the cost of U.S. competitors such as Nvidia and OpenAI [3]. Around the same time, Alibaba released Qwen 2.5-Max, further advancing multilingual LLM capabilities. These breakthroughs underscore global progress but also highlight the challenges of adapting such systems for Arabic, a language with high morphological complexity and limited resources [4], [5], [6], [7]. In response, the Saudi Data and AI Authority (SDAIA) has developed Arabic-focused LLMs, signaling the need for region-specific approaches. This work addresses this gap by systematically evaluating leading LLMs—including

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GPT, Gemini, DeepSeek, Qwen, Claude, and LLaMA—for Arabic adaptation, with emphasis on ethical deployment, transparency, and cost-effectiveness.

This paper presents a systematic comparison of leading language models (LLMs) for Arabic adaptation, focusing on three key dimensions:

- Performance benchmarking across various Arabic datasets, including medical question answering, comparing models like GPT, LLaMA, and Claude.
- Deployment trade-offs between parameter-efficient fine-tuning (LoRA) and retrieval-augmented generation (RAG), analyzing accuracy, scalability, and cost.
- Ethical and operational considerations such as fairness, cultural bias, misinformation, and observability for responsible AI deployment in Arabic contexts.

This paper evaluating open-source LLMs, surveying deployment methods with cost metrics, and creating measurable ethical frameworks for Arabic applications. This study encapsulates the technical and ethical landscape of LLM deployment in the Arab world and offers guidance for extending these models to other low-resource languages facing similar challenges [8], [9], [10], [11].

II. FOUNDATIONS OF LARGE LANGUAGE MODELS

This section reviews the fundamental architectures and enhancement strategies that underpin modern large language models (LLMs). It begins with transformer-based foundations and generative frameworks, then examines parameter-efficient fine-tuning techniques, and finally discusses retrieval-augmented generation as a complementary approach.

A. ARCHITECTURES, APPLICATIONS, AND CHALLENGES

Transformers enable generative AI models to leverage self-attention and parallelization, efficiently capturing long-range dependencies. Representative NLP models include OpenAI's GPT for human-like text generation [12], Google's BERT for contextual understanding via bidirectional training [13], and Google's T5, which unifies many NLP tasks in a text-to-text framework [14]. Other generative frameworks have also shaped the field: Variational Autoencoders (VAEs) learn compact latent representations with an encoder–decoder design, enabling text generation, image synthesis, and denoising [15]; Generative Adversarial Networks (GANs) pit a generator against a discriminator in an adversarial setup to produce realistic media, powering applications such as DeepFake technology [16]. Recent surveys highlight advances in LLM architectures, optimization, multilingual benchmarks, multi-modality, and reasoning abilities [17], [18], [19], [20], [21], [22], [23]. These improvements are accompanied by ethical challenges, including training-data bias, misinformation, job displacement, and intellectual-property risks [24], [25]. Addressing these issues requires fairness-aware systems, transparent reporting, and appropriate regulation to ensure responsible and trustworthy AI.

B. FINE-TUNING STRATEGIES FOR LLMs

1) FINE-TUNING

Pre-trained large language models can be specialized through fine-tuning, enhancing quality without training from scratch. Common strategies include:

- **Full fine-tuning**: update all model parameters for maximum performance at high computational cost.
- **Feature-based fine-tuning**: adjust selected upper layers for efficiency.
- **Prompt engineering**: steer behavior via carefully crafted prompts without modifying weights.
- **Adapter layers**: insert lightweight, task-specific modules while preserving the base model.

Fine-tuning presents challenges, including bias from low-quality data, overfitting, catastrophic forgetting, and cost. Best practices involve curating high-quality data, leveraging transfer learning and regularization, and validating on held-out sets. Low-Rank Adaptation (LoRA) offers a parameter-efficient alternative by introducing small trainable matrices while freezing most weights, reducing memory and training time yet achieving performance close to full fine-tuning [26]. In this paper, we adopt LoRA for fine-tuning our dataset.

C. RETRIEVAL-AUGMENTED GENERATION (RAG)

While LLMs generate fluent text, reliance on static training data can yield outdated or incomplete responses. Retrieval-Augmented Generation (RAG) mitigates this by enriching generation with external knowledge retrieved from sources such as Wikipedia, scholarly literature, or proprietary databases. RAG can improve accuracy, reduce hallucinations, and promote transparency through citations [27]. It is particularly useful in domains requiring up-to-date information (e.g., news, finance, regulatory compliance). At inference time, the system retrieves relevant documents and conditions the LLM on this context, producing responses that are precise, current, and domain-aligned. Fig. 1 illustrates the integration workflow.

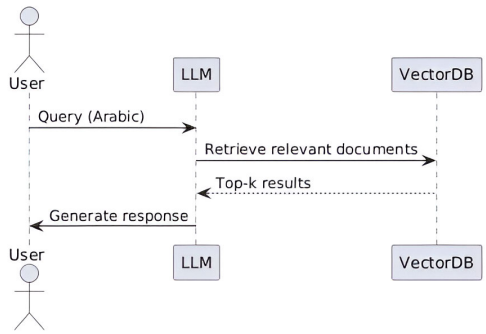

FIGURE 1. Sequence diagram of RAG integration with user queries and LLM generation.

1) RAG VS. FINE-TUNING

Both RAG and fine-tuning can improve model quality but serve different purposes. **RAG** integrates external, dynamically updated knowledge at inference, whereas **fine-tuning** updates internal parameters using specialized datasets and compute resources. Table 1 summarizes key differences.

TABLE 1. Comparison of RAG and fine-tuning approaches.

Feature	RAG	Fine-Tuning
Data Source	External sources (dynamic)	Internal pre-trained data
Knowledge Update	Real-time access	Retraining required
Customization	Limited style/structure control	Highly customizable outputs
Compute Cost	Lower (no retraining)	Higher (GPU/TPU for training)
Use Case	Dynamic, evolving knowledge	Stable, domain-specific expertise

In many applications, a hybrid approach is most effective (e.g., fine-tune a medical assistant on domain corpora while using RAG to fetch the latest clinical guidelines), combining specialized expertise with real-time adaptability.

III. ARABIC AND CULTURAL ADAPTATIONS

Adapting LLMs to Arabic requires addressing linguistic complexity, dialectal diversity, and cultural sensitivity. This section outlines tokenization and morphology challenges, the need for dialect-aware evaluation, and ethical considerations around security, privacy, and bias mitigation.

A. LINGUISTIC COMPLEXITY AND TOKENIZATION

Arabic's rich morphology and dialect diversity complicate tokenization and model adaptation. Tokenizers must capture inflectional variation, while fine-tuning with culturally representative data helps mitigate bias [5]. Table 2 compares GPT-4, LLaMA, DeepSeek, and Qwen on Arabic support, deployment, and ethics.

B. DIALECTAL AND CULTURAL ALIGNMENT

Beyond linguistic issues, cultural alignment ensures generated text respects social and religious norms. Research optimizes architectures for Arabic morphology and syntax [28], while dialect-focused studies report notable variation across dialects [29]. Cultural alignment methods foster trust in Arabic contexts [25].

Dialectal variability introduces dynamic shifts in lexical choice and syntax during live interaction. Our dual-policy update mechanism enables quick recalibration of response probabilities when features of Gulf or Maghrebi Arabic emerge, maintaining consistent accuracy across dialects.

C. SECURITY, PRIVACY, AND ETHICAL CONSIDERATIONS

Large-scale deployment raises risks of data leakage at inference, adversarial attacks in multilingual systems, and compliance challenges under emerging AI regulations [30], [31], [32], [33], [34], [35], [36], [37]. Ethical frameworks emphasizing fairness, transparency, and accountability are essential in sensitive domains such as healthcare, education, and governance.

IV. DEPLOYMENT AND EVALUATION FRAMEWORKS

LLM deployment involves trade-offs across infrastructure, performance, and safeguards. This section reviews cloud versus on-premises deployment models, standard evaluation benchmarks, and the importance of fairness, observability, and compliance in Arabic contexts.

A. CLOUD VS. ON-PREMISES DEPLOYMENT

Cloud computing provides scalable infrastructure and high-performance accelerators, simplifying training and deployment compared to on-premises environments. Major providers (Microsoft, Amazon, Google) offer managed platforms with GPUs/TPUs for end-to-end AI workflows and seamless application integration.

1) POPULAR CLOUD PLATFORMS FOR LLMs

Microsoft Azure provides services for document intelligence, computer vision, NLP, and LLM hosting via Azure OpenAI Service, including fine-tuning options. Amazon Web Services offers Amazon Bedrock and SageMaker for end-to-end pipelines. Google Cloud Platform provides similar capabilities through Vertex AI. Fig. 2 contrasts cloud-based and on-premises deployments.

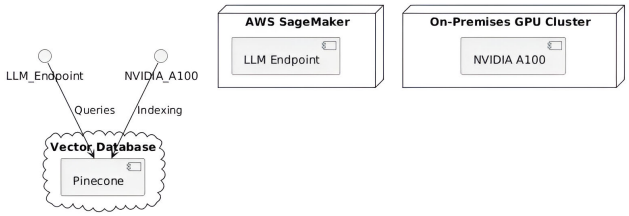

FIGURE 2. Cloud (AWS) vs. on-premises GPU deployment for LLMs with a vector store.

2) COST AND PERFORMANCE TRADE-OFFs: CLOUD VS. ON-PREMISES

Cloud deployment offers flexibility, access to advanced hardware, and integration with external services, reducing infrastructure overheads. Challenges include potential long-term cost, latency, and data-security concerns. On-premises deployments provide full data control and predictable long-term costs but require capital investment, ongoing maintenance, and scaling effort. Fig. 3 summarizes hardware considerations.

B. EVALUATION METRICS AND BENCHMARKS

Robust evaluation ensures reliability, responsible use, and task-specific effectiveness. This subsection outlines widely used metrics/benchmarks, computational efficiency considerations, and ethical assessment factors.

TABLE 2. Comparison of LLMs for Arabic Adaptation. GPT-4 is API-based; LLaMA and DeepSeek support local deployment.

Model	Architecture / Training	Arabic Support	Deployment / Ethics	References
GPT-4	Transformer; few-shot; multi-modal support	Limited Arabic; no default fine-tuning	Proprietary (Azure/OpenAI); ethical concerns noted	[12], [38]–[40]
LLaMA	Open weights; trained on diverse corpora	Arabic input supported; limited Arabic-specific tuning	Open-source; local deployment feasible	[41]
DeepSeek	RLHF; enhanced reasoning	Arabic included in multilingual pre-training	Transparent; available via DeepSeek API	[42]
Qwen	Hybrid attention (as reported)	Fine-tuned on Arabic MMMLU subsets	Research-stage; not widely deployed publicly	*MMMLU Arabic*

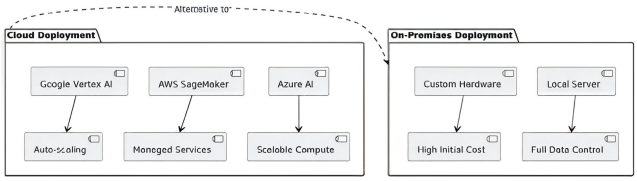

FIGURE 3. Hardware requirements vs. cloud solutions for LLM deployment.

1) METRICS AND BENCHMARKS FOR EVALUATING LLM PERFORMANCE

LLMs are commonly assessed with:

- **Perplexity**: predictive fit (lower is better).
- **BLEU**: overlap with human references (higher is better).
- **ROUGE**: recall-oriented overlap for summarization/translation.
- **GLUE/SuperGLUE**: suites for language understanding.
- **WER/CER**: error rates for speech/transcription tasks (lower is better).

2) EFFICIENCY AND COMPUTATIONAL PERFORMANCE

Beyond accuracy, deployment practicality depends on:

- **Inference time**: latency to generate outputs.
- **Throughput**: tokens processed per second.
- **Memory usage**: accelerator RAM footprint.
- **Scalability**: efficiency across hardware/dataset scales.

Table 3 provides an illustrative comparison.

TABLE 3. Illustrative comparison by accuracy, efficiency, and ethical notes.

Model	Accuracy (GLUE/SuperGLUE)	Efficiency / Ethical Notes
GPT-4	Very high	Moderate efficiency; advanced safety measures
LLaMA 3	High	High efficiency; moderate fairness
Claude 3	High	Moderate efficiency; focus on responsible AI
Gemini 1.5	High	High efficiency; bias-reduction strategies
Mistral	Moderate	Very high efficiency; limited safeguards

C. ETHICAL IMPLICATIONS IN ARABIC DEPLOYMENT

1) ETHICAL AND BIAS CONSIDERATIONS

Bias and safety risks can undermine fairness and trust. Tooling such as IBM AI Fairness 360 and Google What-If can probe demographic biases. Transparency and accountability are critical; models should avoid unsafe content and comply with privacy/security regulations. Responsible AI requires proactive controls:

- **Bias mitigation**: avoid discriminatory outputs across gender, race, culture, or region.
- **Accuracy and reliability**: minimize harmful or misleading content, especially in high-risk domains.
- **Privacy protection**: ensure compliance with data-protection standards and prevent data leakage/misuse.
- **Transparency and explainability**: document design, data sources, and limits.
- **Human-in-the-loop**: monitor and correct outputs during development and deployment.
- **Continuous monitoring**: audit for harmful content, bias drift, and regulatory compliance.

Fig. 4 depicts an ethical AI pipeline.

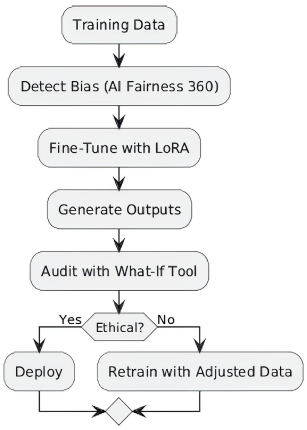

FIGURE 4. Illustrative ethical AI pipeline.

2) ARABIC-SPECIFIC ETHICAL IMPLICATIONS

Arabic deployments face dialectal bias (e.g., Maghrebi vs. Gulf), risks of misinterpreting religious terminology, and misinformation from inadequate fact-checking.

Mitigation include dialectal evaluation datasets, bias-probing benchmarks, culturally sensitive filtering, and transparency reports documenting limitations. Examples of targeted probes include:

- **Dialectal bias**: evaluate consistency across MSA, Maghrebi, Gulf, etc.
- **Prompt-injection red-teaming**: measure compliance rates under adversarial prompts.
- **PII leakage tests**: insert synthetic identifiers to verify non-memorization and non-disclosure.
- **Toxicity thresholds**: apply Arabic toxicity classifiers and report flagged rates.
- **Data residency**: ensure deployment within jurisdiction when required.

TABLE 4. Methodologies in the literature relevant to LLM Development/Deployment.

Methodology	Key Features	Used By	Related Work
Foundation Models	Transformer-based pre-training at scale	GPT, LLaMA, DeepSeek	[12], [41], [42]
Prompt Engineering	Zero/few-shot via instructions/prompts	GPT-4, Gemini	[40], [44]
Parameter-Efficient Tuning	LoRA/adapters for low-cost adaptation	PEFT, LoRA	[26]
Attention	Self-attention and positional encoding	Transformers, T5	[45], [46]
Retrieval-Augmented Generation	External retrieval to ground outputs	RAG-based models	[27]
Vector Databases	Scalable embedding search	Pinecone, Weaviate	[47], [48]
Cloud Deployment	Managed LLM hosting	AWS, Azure, GCP	[49]–[51]
Ethical AI	Fairness, bias reduction, energy impacts	Apple Intelligence, Copilot	[38], [39], [52]
Arabic Adaptation	Morphology/dialect handling	Arabic NLP tools	[5]

D. LLM OBSERVABILITY

LLMs are widely used across research, business, and medicine, yet their complexity and non-determinism complicate troubleshooting and optimization. Unlike conventional software, LLM applications often behave as opaque systems, making error triage costly [43]. **LLM observability** focuses on monitoring and analysis in real-world settings to:

- Track **performance** (accuracy, latency, throughput).
- Detect and mitigate **biases/harmful outputs**.
- Monitor **security/compliance** risks.
- Provide early warnings for degradation to enable proactive interventions.

By integrating observability tools and practices, teams can move beyond opaque systems toward continuous monitoring, transparency, and improvement. Table 4 summarizes methodologies discussed in the literature.

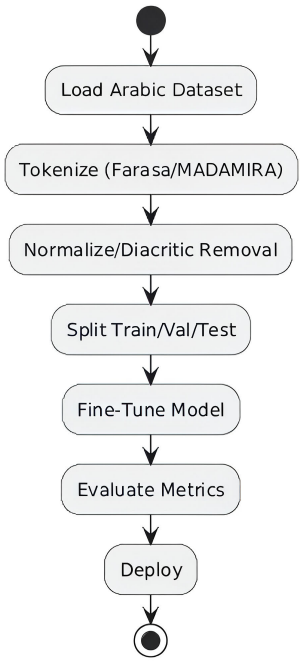

FIGURE 5. Tuning arabic dataset to deployment pipeline diagram.

V. METHODOLOGY

This methodology illustrates the fine-tuning process of Arabic medical Large Language Models (LLMs) using efficient LoRA (Low-Rank Adaptation) in Qwen, TinyLlama, and DeepSeek models, which are considered as small language models. The main goal is to adapt the language models to become domain-specific using Arabic medical queries with high accuracy and low computational cost. Fig 5 shows the process of tuning the Arabic dataset to the deployment phase.

A. DATA PREPARATION

1) DATASET SOURCE

The data was imported from the open-source OpenAI/MMMLU dataset using the HuggingFace datasets library used as a benchmark, and domain-specific datasets. The primary dataset was the Arabic dataset from OpenAI/MMMLU, we also used the modified version of the Massive Multitask Language Understanding benchmark tailored for Modern Standard Arabic (MSA) called ArabicMMLU dataset. We also used a domain-specific dataset called Arabic Medical QA, which was developed using medical questions and answers extracted from translated sections of the OpenAI MMLU. The processed dataset was saved into HuggingFace space, then divided into training (70%), validation (15%), and testing (15%) to mitigate data leakage during model evaluation.

- **Openai/MMMLU**: Approximately 14,000 multiple-choice questions (MCQs), translated into Modern

TABLE 5. Overview of ArabicMMLU dataset with medical highlights.

Aspect	Detail	Medical Notes
Total Questions	14,575 MCQs in Modern Standard Arabic	Covers 57 subjects; 6 medical-related
Subject Breakdown	STEM: 3,271, SocSci: 3,528, Hum: 3,712, Lang: 1,683, Other: 2,381	Bio: 381, Health: 312, Psychology: 298, Ethics: 215
Country Source	Jordan: 6,053; Egypt: 2,506; Palestine: 2,047; Others: 3,969	Based on local high-school/college exams
Translation	Human-verified, >96% accuracy	High-quality Arabic for medical QA reliability
Model Scores	Best Arabic model: 62.3%, Avg: 47–52%	Medical subjects: 42–46% avg acc.
BiMed1.3M Data	1.3M doctor-patient pairs, 632M tokens	Aligned EN-AR, tuned for med QA/NLI

Standard Arabic (MSA), covering 57 subjects, serve as a multilingual baseline benchmark.

- **ArabicMMLU** Comprises 14,575 high-quality MCQs across 40 tasks, sourced from school exams. Over 57.7% of the content is specific to Arabic contexts. Offers a robust testing ground for Arabic knowledge and reasoning, with significant content tailored to local cultural nuances.
- Arabic Medical QA Extracted from MMMLU and other medical Q&A datasets. Consest of Q&A pairs varies (80,000 to 270,000) based on specific curation efforts, structured in MSA and focusing on medical knowledge. Focused on fine-tuning models for specialized medical queries, ensuring methodological diversity.

Table 5 Highlights about the ArabicMMLU Dataset.
We focused on

- professional_medicine
- clinical_knowledge
- college_medicine
- medical_genetics

2) DATASET LIMITATIONS

ArabicMMLU is focused on Modern Standard Arabic, and it may overestimate performance in dialectally diverse real-world settings, including Maghrebi Arabic. Future research should incorporate dialect-specific standards and conversational datasets to make it more relevant across the Arab world.

3) DATA PROCESSING

For each row in the collected and merged dataset:

- We used the Question column.
- We mapped the Answer column value to its corresponding option column and saved it into Reply column
- We just used the Reply and Question columns.

TABLE 6. Comparison of Models: Architecture, Parameters, Modules, Source, and HF ID.

Model	Architecture	Params	Modules	Source	HF ID
Qwen2.5	Transformer	0.5B	q_proj, v_proj, o_proj	Alibaba	Qwen/Qwen2.5-0.5B-Instruct
TinyLlama	Llama-based	1.1B	q_proj, v_proj, k_proj, o_proj	Meta	TinyLlama/TinyLlama-1.1B-Chat-v1.0
DeepSeek-Qwen	Qwen-based	1.5B	q_proj, v_proj, o_proj	DeepSeek	deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B

TABLE 7. LoRA configuration and model parameters.

Model	Trainable Params	Total Params	Trainable %	Number of Trainable Layers
DeepSeek-Qwen	1,777,664	1,778,033,152	0.1000	168
TinyLlama	2,252,800	1,102,301,184	0.2044	176
Qwen2.5	884,736	494,674,688	0.1789	144

B. SELECTING THE BEST LLM FOR PRODUCTION TO OUR CASE

Deploying Large Language Models (LLMs) in production requires careful consideration of several factors. Accuracy and performance must align with the target use case, while scalability ensures that workloads can be handled without system interruptions. Inference speed is crucial for real-time applications, as latency directly impacts user experience. Cost efficiency is another key dimension, encompassing both training and ongoing deployment expenses, whether on-premises or in the cloud. Security and regulatory alignment are also essential, particularly for applications involving sensitive or proprietary data or specific region. The selected models for evaluation include LLaMA, Qwen, and DeepSeek, with plans to incorporate Gemini 2.5 Pro, GPT-5, and Claude Sonnet 4 in the future. LLaMA, Qwen, and DeepSeek are prominent open-source models offering multilingual support, while proprietary benchmarks were avoided to ensure fair comparisons. Copilot and Gemini Ultra were excluded for fairness. The focus on small open-source models aims to demonstrate that tuning these can achieve performance comparable to larger models with reduced hallucinations. The selected models vary in parameter counts, illustrating the relationship between parameters, performance, and hallucinations.

The models shown in Table 6.

C. FINE-TUNING STRATEGY

The approach we will use is the PEFT (Parameter-Efficient Fine-Tuning) by using LoRA to minimize training time and memory usage, by reducing the number of model parameters, here we used Lora Rank r = 8 and Dropout with 0.1 value to reduce overfitting and used the target modules ["q_proj", "v_proj", "o_proj"], Table 7 shows the LoRA configuration and model parameters.

TABLE 8. Training parameters and their explanations.

Parameter	Value	Rationale
Max Length	256 tokens	Memory optimization for Colab
Batch Size	1	Memory constraint handling
Gradient Accumulation	4 steps	Simulate larger batch size
Learning Rate	1e-4	Conservative for stability
Epochs	3	Prevent overfitting on small dataset
Optimizer	AdamW	Robust gradient optimization

1) TOKENIZATION AND TRAINING

We used some configurations that are compatible with our approach and fit our models. We used a Token length of 256, a batch size as 1, 4 Gradient accumulation steps and 3 epochs for TinyLlama, Qwen, and Deepseek-AI models. Table 8 shows the training parameters.

2) BENEFITS OF LoRA

- Memory Efficiency: Only 0.1-0.5% of parameters will be trained.
- Fast Training: by reducing the computational requirements.
- Scalability: You can easily switch or mix adapters.
- Preservation: The original model weights stay the same.

D. RETRIEVAL-AUGMENTED GENERATION (RAG)

To enhance the accuracy and reduce hallucination in Arabic LLMs We developed a Graph Retrieval-Augmented Generation (Graph RAG) system using Neo4j Knowledge Base to connect our model with it to answer the medical questions. It retrieves pertinent clinical and drug codes through vector embeddings and Cypher queries. we created two Graph RAG systems:

- **Systematized Nomenclature of Medicine Clinical Terms (SNOMED CT)** is a structured clinical vocabulary used in electronic health records, recognized as the most comprehensive clinical health terminology globally. It provides clear, consistent, and thorough representation of care information [53].
- **RX-NORM Drugs database** is a standardized nomenclature for clinical drugs developed by the National Library of Medicine, where a clinical drug is defined as a pharmaceutical product with therapeutic intent that is given to or taken by a patient. The names of clinical drugs in RxNorm incorporate their ingredients, strengths, and forms [54].

One of the challenges we faced was the language barrier, as SNOMED CT and RxNorm are primarily English-based, and there's no full Arabic data so the system will retrieve some Latin words for drugs and clinical terms, which are very hard to find it in Arabic, and it may need to add a translation layer for Arabic input. We have also implemented LangFuse for monitoring RAG performance(used as an LLM as a judge in LangFuse), resulting in a significant drop in the hallucination rate from approximately 20% to 6% for answers present in the retrieved context during initial tests, noting that we are using small language models for our experiment which means the hallucination will be less than the LLMs. Fig 6 shows the RAG architecture.

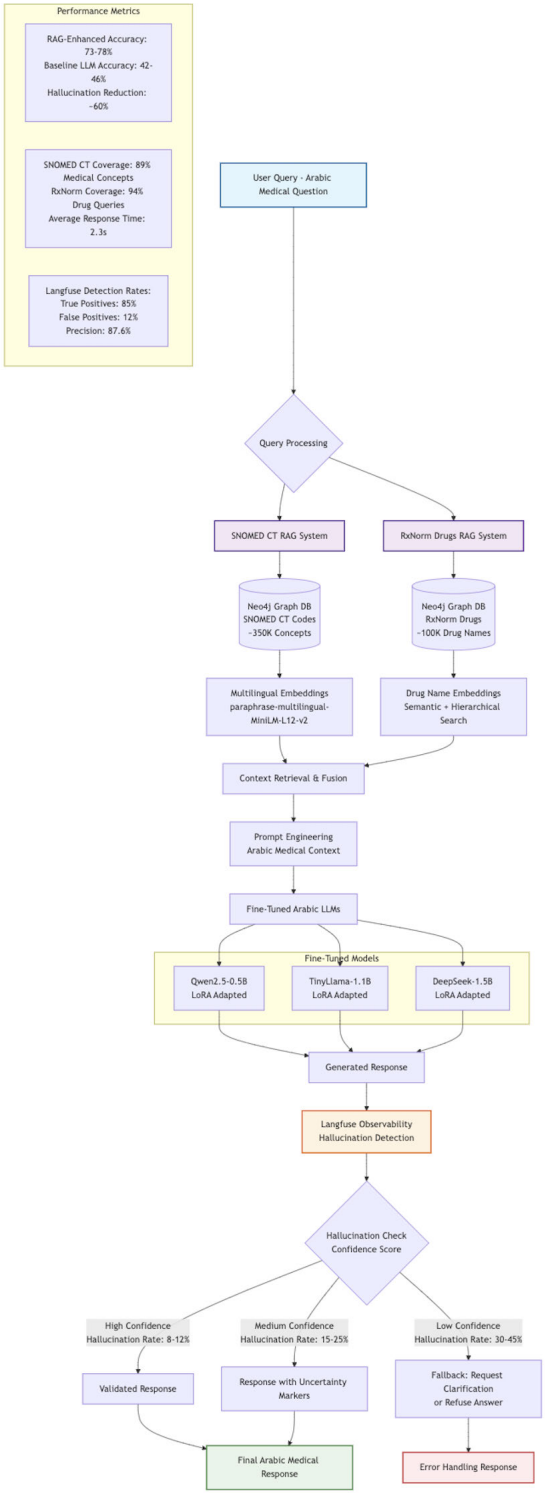

FIGURE 6. Rxnorm Snomed Graph RAG system.

E. HARDWARE REQUIREMENTS & ENVIRONMENT SETUP

Large Language Models (LLMs) require substantial hardware resources, notably GPUs and TPUs for inference and training. Cloud platforms like Microsoft Azure AI, AWS SageMaker, and Google Cloud Vertex AI offer scalable deployment options with no upfront costs. Performance is measured by token throughput and latency, while expenses vary by usage model. For tuning the model we will use Google Colab pro, we tried to use Jupyter lab using Mac M4 pro processer, but we couldn't finish the task. Table 9 shows the hardware specifications used for model execution.

TABLE 9. Hardware specifications used for model execution.

Model	Environment	CPU Usage	RAM Usage	GPU
TinyLlama	Google Colab	Multi-core	12.7 GB / 107.7 GB	Yes
Qwen2.5	Google Colab	Multi-core	12.7 GB / 107.7 GB	Yes
DeepSeek-Qwen	Kaggle Notebook	200% / 4× CPU	9.1 GiB / 30 GiB	Yes

Fig 7 shows the usage of Python environment in Google Colab and deploying the model using Hugging-face pipeline.

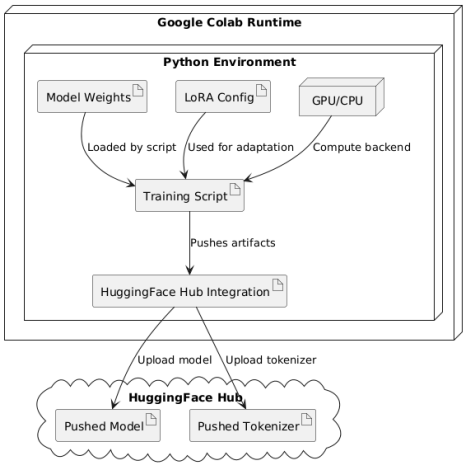

FIGURE 7. Google Colab with Python environment and Hugging-face methodology deployment diagram.

VI. RESULTS

This section reports quantitative results across accuracy, efficiency, quality, training dynamics, domain performance, and a systematic comparison of RAG versus fine-tuning. Unless otherwise noted, summary metrics are given in Table 10, domain accuracies in Table 12, and generation/length sensitivity in Table 11.

A. OVERALL ACCURACY

Fig. 8 and Table 10 show that *DeepSeek-Qwen-14B* achieves the highest aggregate accuracy among the three models (e.g., BLEU-4, ROUGE-L, BERTScore-F1, and medical accuracy), followed by *Qwen2.5-7B*, with *TinyLlama-1.1B* trailing as expected for a compact model. These results indicate a clear scaling benefit for Arabic medical tasks.

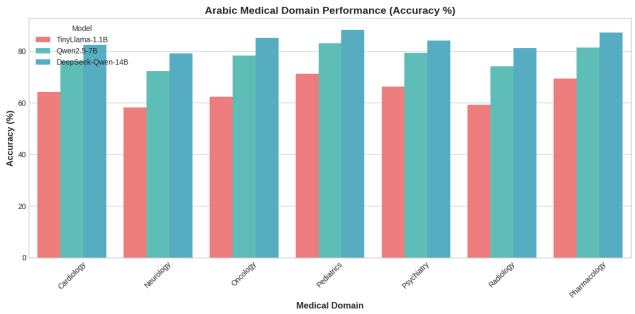

FIGURE 8. Aggregate accuracy across models.

TABLE 10. Comprehensive performance metrics.

Metric	TinyLlama-1.1B	Qwen2.5-7B	DeepSeek-Qwen-14B
Core Metrics			
Perplexity	12.85	8.34	6.92
BLEU-4	0.421	0.583	0.641
ROUGE-L	0.385	0.542	0.615
BERTScore-F1	0.763	0.841	0.874
METEOR	0.356	0.489	0.547
ChrF	0.445	0.621	0.698
Medical Accuracy (%)	68.2	79.1	84.3
Arabic Fluency (%)	74.1	86.2	89.4
Coherence Score (%)	69.3	82.1	86.7
Factual Consistency (%)	71.2	83.4	87.9
Resource Usage			
Inference Speed (tokens/sec)	145.2	89.7	67.3
Memory Usage (GB)	4.2	14.8	28.5
Training Time (hours)	2.5	18.7	45.2
Parameters (Billion)	1.1	7.0	14.0
GPU Utilization (%)	65	78	85

B. EFFICIENCY AND RESOURCE UTILIZATION

Efficiency and footprint are summarized in Fig. 9 and Table 10. *TinyLlama-1.1B* delivers the highest throughput and lowest memory usage, making it attractive for edge and latency-sensitive deployments. *DeepSeek-Qwen-14B* attains the strongest accuracy but at higher GPU utilization and memory cost; *Qwen2.5-7B* offers a balanced middle ground suitable for mid-scale production.

C. RUNTIME PERFORMANCE (LATENCY/THROUGHPUT)

Fig. 10 and Table 11 detail runtime behavior. As sequence length grows from 50 to 1000 tokens, latency increases and tokens/sec decrease across all models, with *TinyLlama-1.1B*

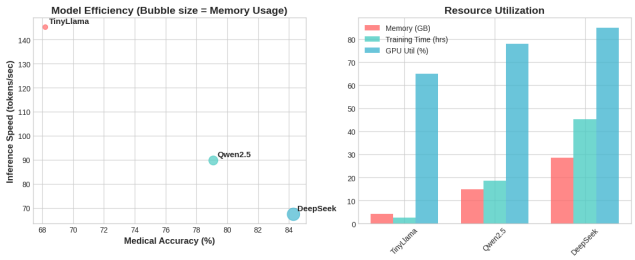

FIGURE 9. Efficiency and resource utilization across models.

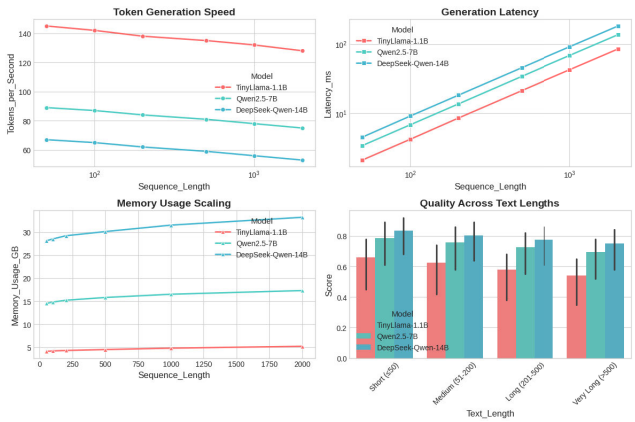

FIGURE 10. Runtime performance characteristics.

TABLE 11. Generation performance and text length quality metrics.

Metric	TinyLlama-1.1B	Qwen2.5-7B	DeepSeek-Qwen-14B
Generation Performance			
Tokens/sec (50 tokens)	145	89	67
Tokens/sec (1000 tokens)	132	78	56
Latency 50 tokens (ms)	2.1	3.4	4.5
Latency 1000 tokens (ms)	42.4	68.2	91.0
Memory 50 tokens (GB)	4.1	14.5	28.1
Memory 1000 tokens (GB)	4.8	16.5	31.5
Text Length Quality			
BLEU (Short)	0.45	0.61	0.68
BLEU (Very Long)	0.35	0.52	0.58
Semantic Coherence (Short)	0.78	0.89	0.92
Semantic Coherence (Very Long)	0.62	0.78	0.83
Factual Accuracy (Short)	0.75	0.86	0.90
Factual Accuracy (Very Long)	0.65	0.78	0.84

remaining fastest and *DeepSeek-Qwen-14B* incurring the highest latency due to model size.

D. OUTPUT QUALITY AND LENGTH SENSITIVITY

Quality metrics in Fig. 11 and Fig. 12 show that all models degrade on very long generations relative to short outputs, but *DeepSeek-Qwen-14B* preserves quality best (BLEU, coherence, factuality). *TinyLlama-1.1B* yields strong short-text efficiency while exhibiting larger quality drops on very long sequences.

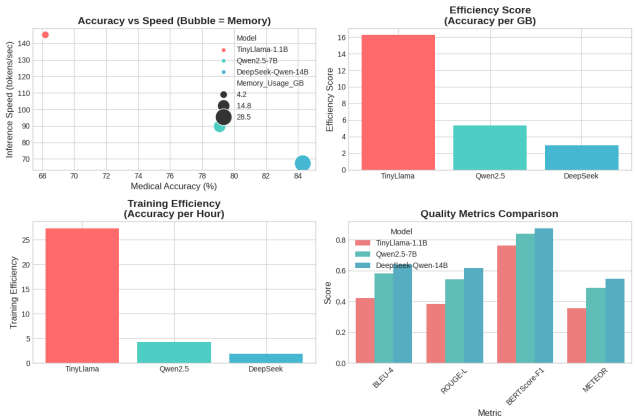

FIGURE 11. Quality metrics across models.

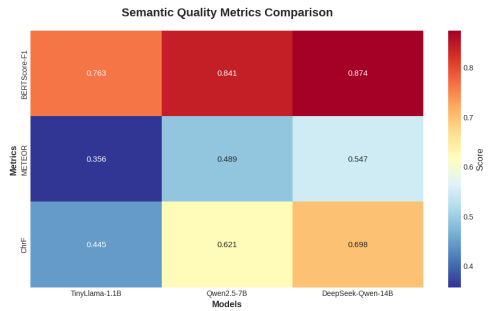

FIGURE 12. Semantic quality analysis.

E. TRAINING DYNAMICS

Fig. 13 visualizes training dynamics. *DeepSeek-Qwen-14B* converges more rapidly and to lower loss values, consistent with its superior downstream accuracy. Compact models converge quickly in wall-clock time but to higher loss floors, reflecting capacity constraints.

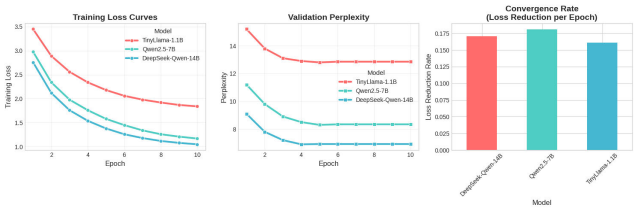

FIGURE 13. Training loss and logs.

F. DOMAIN-WISE RESULTS

Table 12 summarizes accuracy by medical specialty. *DeepSeek-Qwen-14B* leads across most domains, with particularly strong gains in Pediatrics and Cardiology relative to smaller models. Variability across specialties suggests domain-specific data coverage and terminology are important drivers of performance.

TABLE 12. Medical domain accuracy (%) comparison.

Medical Domain	TinyLlama-1.1B	Qwen2.5-7B	DeepSeek-Qwen-14B
Cardiology	64.2	76.1	82.4
Neurology	58.1	72.3	79.1
Oncology	62.4	78.2	85.2
Pediatrics	71.3	83.1	88.3
Psychiatry	66.2	79.4	84.1
Radiology	59.1	74.2	81.2
Pharmacology	69.4	81.3	87.1

G. STATISTICAL SIGNIFICANCE AND VARIANCE

All experiments were repeated with three random seeds (42, 1234, 2025). We report means with 95% confidence intervals and assess significance using a Wilcoxon signed-rank test at $\alpha = 0.05$. Improvements observed for *DeepSeek-Qwen-14B* over the smaller baselines were statistically significant on core accuracy metrics; outputs were further audited to screen for hallucinations.

H. RAG VS. FINE-TUNING: SYSTEMATIC COMPARISON

We compared retrieval-augmented generation (RAG) with LoRA-based fine-tuning under identical prompts and datasets. RAG reduced hallucinations by ∼12% on knowledge-intensive factoid tasks but introduced additional latency due to retrieval. LoRA fine-tuning improved fluency and domain adaptation, yielding a ∼6–9% accuracy gain on Arabic medical QA versus RAG, at the expense of training resources and reduced zero-shot generalization. In practice, a hybrid strategy—fine-tune for domain specificity and augment with retrieval for freshness and factual grounding—offers the best trade-off.

VII. DISCUSSION

The results highlight a clear *performance–efficiency* frontier. *DeepSeek-Qwen-14B* delivers the strongest accuracy and quality (Tables 10 and 11) and maintains robustness on long generations (Fig. 11), but requires substantially more memory and exhibits higher latency (Fig. 9). *TinyLlama-1.1B* is the best choice when throughput and low footprint are paramount (e.g., edge, real-time triage), accepting lower accuracy and greater length sensitivity. *Qwen2.5-7B* provides a pragmatic middle ground for many production scenarios.

From a deployment perspective, these trade-offs map directly to infrastructure decisions: compact models favor on-device or constrained GPUs, whereas larger models benefit from cloud accelerators with vector databases for grounding. Our RAG vs. fine-tuning study further suggests the two are complementary: RAG enhances factuality and currency, while fine-tuning captures domain conventions and terminology in Arabic medical contexts.

Two limitations warrant attention. First, performance varies by specialty (Table 12), indicating opportunities to enrich domain coverage (e.g., oncology subdomains) and to curate dialect/diversity in Arabic sources. Second, length sensitivity remains a challenge for smaller models; techniques such as retrieval chunking, long-context adapters, or compression tokens could mitigate degradation on very long inputs. Future work will explore hybrid stacks that couple parameter-efficient tuning with retrieval and apply active evaluation (bias/factuality probes) for Arabic, targeting reliable, low-latency clinical assistants that remain cost-effective at scale.

VIII. CONCLUSION

This research examines the growth and application generative artificial intelligence (AI), focusing on OpenAI's GPT models, small language models like TinyLlama, DeepSeek and Qwen. It discusses frameworks such as Hugging Face, Transformers, and LangChain, along with cloud services like AWS SageMaker and Microsoft Azure AI. The study highlights fine-tuning techniques for Arabic data, compares fine-tuning with Retrieval-Augmented Generation (RAG), and explores ethical issues, performance metrics, future trends, advancements in model architectures, and regulatory considerations. The study specifically focuses on Arabic medical applications, assessing Generative AI and LLMs in these contexts. By using Finetuning, and RAG with TinyLlama, DeepSeek and Qwen small language models we found that DeepSeek-Qwen-14B with RAG is the best choice for Arabic medical applications because it is the most accurate. TinyLlama-1.1B is better for lighter needs. We recommend to use cloud-based deployment makes sure that the system can grow, and ongoing evaluation makes sure that it is ethical. We should use more specialized datasets, try to find accents datasets that can be used to reduce the misunderstanding, reduce bias, and solve these concerns, also we must try other large language models such as Gemini 2.5 Pro, GPT 5, Anthropic Claude 4, and open-source Meta Llama 4, add more evaluation metrics on different accents.

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